

Core Pure Mathematics 2

Notes · Edexcel A Level Further Mathematics — Core Pure 2 · Chapter 3: Methods in Calculus

Chapter 3: Methods in Calculus

Why this matters

A Level Maths covers standard definite integrals where the function is well-behaved on a closed interval. This chapter extends integration to cases where something breaks down — an infinite interval, or a function that shoots off to infinity within the interval — via **improper integrals**. It also introduces two genuinely new applications of integration: finding the **mean value** of a function, and finding the **arc length** of a curve or the **surface area** generated when a curve is rotated. These are core techniques you'll reuse throughout further maths and in mechanics/further mechanics.

Concept: improper integrals

An integral is **improper** if either:

1. one or both limits of integration are infinite, or
2. the function is undefined (has a discontinuity, often a vertical asymptote) at a point within or at an endpoint of the interval.

We handle these by replacing the problematic limit with a variable (say t), evaluating the definite integral, then taking the limit as t tends to the problem value.

Key vocabulary

Term	Meaning
Improper integral	A definite integral with an infinite limit or a discontinuity in the range
Convergent	The limit defining the improper integral exists (is finite)
Divergent	The limit does not exist or is infinite

Worked example 1 — Improper integral with an infinite limit

WORKED EXAMPLE 1

Question. Determine whether the integral from 1 to ∞ of $1/x^2$ dx converges, and if so, find its value.

Step 1. Replace ∞ with t and write as a limit: **(M1)** the integral from 1 to ∞ of $1/x^2$ dx = $\lim(t \rightarrow \infty)$ [the integral from 1 to t of x^{-2} dx]

Step 2. Integrate: $\int x^{-2} dx = -x^{-1} + c$, so **(M1)** the integral from 1 to t of x^{-2} dx = $[-1/x]$ from 1 to t = $(-1/t) - (-1/1) = 1 - 1/t$

Step 3. Take the limit as $t \rightarrow \infty$: $1/t \rightarrow 0$, so the expression $\rightarrow 1$. **(A1)**

Answer: The integral converges to 1.

EXAM TIP

It is tempting to think "integrating to infinity" is automatically meaningless. The key intuition: as long as the *area under the curve* stops growing fast enough (here, $1/x^2$ shrinks quickly), the total area can settle to a finite number even over an infinite strip. Compare with $\int_1^\infty 1/x \, dx$ (which diverges, giving $\ln t \rightarrow \infty$) to show that "shrinking to zero" isn't enough on its own — the *rate* matters.

Worked example 2 — Improper integral with a discontinuity in the interval

WORKED EXAMPLE 2

Question. Show that the integral from 0 to 4 of $1/\sqrt{x} \, dx$ converges, and find its value.

Step 1. $1/\sqrt{x}$ is undefined at $x = 0$, which is the lower limit, so this is improper. Replace 0 with t : **(M1)** the integral from 0 to 4 of $x^{(-1/2)} \, dx = \lim(t \rightarrow 0^+) [\text{the integral from } t \text{ to } 4 \text{ of } x^{(-1/2)} \, dx]$

Step 2. Integrate: $\int x^{(-1/2)} \, dx = 2x^{(1/2)} + c$ **(M1)** the integral from t to 4 of $x^{(-1/2)} \, dx = [2\sqrt{x}]$ from t to 4 $= 2\sqrt{4} - 2\sqrt{t} = 4 - 2\sqrt{t}$

Step 3. Take the limit as $t \rightarrow 0^+$: $2\sqrt{t} \rightarrow 0$, so the expression $\rightarrow 4$. **(A1)**

Answer: The integral converges to 4.

EXAM TIP

Always identify where the improper behaviour is (which limit, and why) before setting up the limit — jumping straight to integrating without stating the issue costs the method mark even if the final number is right, because the examiner is checking you understand why a limit process is needed at all.

What divergence actually looks like: $\int_1^\infty (1/x) \, dx = \lim(t \rightarrow \infty) [\ln x]_1^t = \lim(t \rightarrow \infty) \ln t$, and $\ln t \rightarrow \infty$ as $t \rightarrow \infty$, so this integral **diverges** — there is no finite value to report, and "the integral diverges" is itself a complete, correct final answer (don't try to force a number out of it).

Concept: mean value of a function

The mean (average) value of a function $f(x)$ over the interval $[a, b]$ is defined as:

$$\text{Mean value} = (1/(b - a)) \times \text{the integral from } a \text{ to } b \text{ of } f(x) \, dx$$

This generalises the idea of an average of discrete values to a continuous function — it is the height of the rectangle over $[a, b]$ with the same area as the region under the curve.

Key formula

KEY FORMULA

$$\text{Mean value of } f(x) \text{ on } [a, b] = (1/(b-a)) \int_a^b f(x) \, dx$$

Worked example 3 — Finding a mean value

WORKED EXAMPLE 3

Question. Find the mean value of $f(x) = \sin x$ over the interval $[0, \pi]$.

Step 1. Apply the formula with $a = 0$, $b = \pi$: **(M1)** Mean value $= (1/(\pi - 0)) \times$ the integral from 0 to π of $\sin x \, dx$

Step 2. Integrate: $\int \sin x \, dx = -\cos x + c$ **(M1)** the integral from 0 to π of $\sin x \, dx = [-\cos x]$ from 0 to $\pi = (-\cos \pi) - (-\cos 0) = (1) - (-1) = 2$

Step 3. Substitute back: **(A1)** Mean value $= (1/\pi) \times 2 = 2/\pi$

Answer: Mean value $= 2/\pi \approx 0.637$.

EXAM TIP

A nice sanity check you can use every time: the mean value must lie between the minimum and maximum of f on the interval. Here $\sin x$ ranges from 0 to 1 on $[0, \pi]$, and $2/\pi \approx 0.637$ sits comfortably inside that range — a quick way to catch arithmetic slips.

Concept: arc length of a curve

For a curve $y = f(x)$ between $x = a$ and $x = b$, the arc length is found by summing infinitesimal straight-line segments using Pythagoras, then taking the limit — this gives the arc length integral:

$$\text{Arc length} = \text{the integral from } a \text{ to } b \text{ of } \sqrt{1 + (dy/dx)^2} \, dx$$

For a curve given parametrically by $x = x(t)$, $y = y(t)$ between $t = t_1$ and $t = t_2$:

$$\text{Arc length} = \text{the integral from } t_1 \text{ to } t_2 \text{ of } \sqrt{(dx/dt)^2 + (dy/dt)^2} \, dt$$

Quick sanity check. For the circle $x = r \cos t$, $y = r \sin t$, $t \in [0, 2\pi]$: $dx/dt = -r \sin t$, $dy/dt = r \cos t$, so $(dx/dt)^2 + (dy/dt)^2 = r^2 \sin^2 t + r^2 \cos^2 t = r^2$. The arc length is then $\int_0^{2\pi} r \, dt = 2\pi r$ — exactly the circumference formula you already know. Whenever you derive a parametric (or Cartesian) arc length integral, it's worth checking it reduces correctly on a shape you can already verify by eye.

Worked example 4 — Arc length of a Cartesian curve

WORKED EXAMPLE 4

Question. Find the length of the curve $y = x^{3/2}$ from $x = 0$ to $x = 5$, giving your answer to 3 significant figures.

Step 1. Differentiate: $dy/dx = (3/2)x^{1/2}$, so $(dy/dx)^2 = (9/4)x$. **(M1)**

Step 2. Set up the arc length integral: **(M1)** Arc length $=$ the integral from 0 to 5 of $\sqrt{1 + (9/4)x}$ dx

Step 3. Integrate using reverse chain rule (let $u = 1 + (9/4)x$, so $du/dx = 9/4$): **(M1)** raising the power by 1 and dividing by the new power gives $\int u^{1/2} \, du = (2/3)u^{3/2} + c$; dividing by $du/dx = 9/4$ (i.e. multiplying by $4/9$) accounts for the chain rule, so $\int (1 + (9/4)x)^{1/2} \, dx = (4/9) \times (2/3) \times (1 + (9/4)x)^{3/2} + c = (8/27)(1 + (9/4)x)^{3/2} + c$

Step 4. Evaluate between the limits: **(A1)(A1)** $[(8/27)(1 + (9/4)x)^{3/2}]$ from 0 to 5 $= (8/27)(1 +$

$$45/4)^{3/2} - (8/27)(1)^{3/2} = (8/27)(49/4)^{3/2} - 8/27 = (8/27)(343/8) - 8/27 = 343/27 - 8/27 = 335/27$$

Answer: Arc length = $335/27 \approx 12.4$ (3 s.f.)

Concept: surface area of revolution

When a curve is rotated about the x-axis (through 2π radians), the surface generated has area given by treating each infinitesimal arc length as a thin circular band of circumference $2\pi y$:

Surface area (rotation about x-axis) = the integral from a to b of $2\pi y \sqrt{1 + (dy/dx)^2} dx$

For rotation about the y-axis, use $2\pi x$ instead of $2\pi y$ (and integrate with respect to y where convenient), and for parametric curves substitute the parametric arc-length integrand:

Surface area (parametric, about x-axis) = the integral from t_1 to t_2 of $2\pi y \sqrt{((dx/dt)^2 + (dy/dt)^2)} dt$

Worked example 5 — Surface area of revolution

WORKED EXAMPLE 5

Question. The curve $y = \sqrt{x}$ for $1 \leq x \leq 4$ is rotated 2π radians about the x-axis. Find the exact surface area generated.

Step 1. Differentiate: $y = x^{1/2}$, $dy/dx = \frac{1}{2}x^{-1/2}$, so $(dy/dx)^2 = 1/(4x)$. **(M1)**

Step 2. Set up the surface area integral: **(M1)** $S = \text{the integral from 1 to 4 of } 2\pi\sqrt{x} \cdot \sqrt{1 + 1/(4x)} dx$

Step 3. Simplify inside the square root: **(M1) — combining terms under a common denominator** $\sqrt{x} \cdot \sqrt{1 + 1/(4x)} = \sqrt{(x + 1/4)} = \sqrt{(4x+1)/4} = \frac{1}{2}\sqrt{4x+1}$ So $S = \text{the integral from 1 to 4 of } 2\pi \cdot \frac{1}{2}\sqrt{4x+1} dx = \pi \int_1^4 \sqrt{4x+1} dx$

Step 4. Integrate: $\int (4x+1)^{1/2} dx = (1/4)(2/3)(4x+1)^{3/2} + c = (1/6)(4x+1)^{3/2} + c$ **(M1)**

Step 5. Evaluate between limits: **(A1)(A1)** $\pi[(1/6)(4x+1)^{3/2}] \text{ from 1 to 4} = \pi[(1/6)(17)^{3/2} - (1/6)(5)^{3/2}] = (\pi/6)(17\sqrt{17} - 5\sqrt{5})$

Answer: $S = (\pi/6)(17\sqrt{17} - 5\sqrt{5})$, exact.

EXAM TIP

It is easy to forget the factor of 2π (writing just $\int y\sqrt{1+(dy/dx)^2}dx$, which is actually a different, non-standard quantity) or use x instead of y (or vice versa) for the wrong axis of rotation. Say out loud "radius is the distance to the axis I'm rotating about" before setting up every surface area integral — this one sentence prevents both errors.

Chapter summary

Idea	Key point
Improper integral	Replace the problem limit with t , integrate, then take the limit as $t \rightarrow$

	the value
Convergence	The integral converges if the limit exists and is finite; otherwise it diverges
Mean value	$(1/(b-a)) \int_a^b f(x) dx$ — average height of the curve over $[a, b]$
Arc length (Cartesian)	$\int_a^b \sqrt{1 + (dy/dx)^2} dx$
Arc length (parametric)	$\int \sqrt{(dx/dt)^2 + (dy/dt)^2} dt$
Surface area of revolution (about x-axis)	$\int 2\pi y \sqrt{1 + (dy/dx)^2} dx$ (use $2\pi x$ for rotation about y-axis)

Common exam mistakes

- Not identifying which limit or point makes an integral improper before starting — always state it explicitly.
- Forgetting to take the limit at the end (leaving the answer in terms of t).
- Dividing by $(b - a)$ incorrectly, or forgetting the $1/(b-a)$ factor entirely, in mean value questions.
- Missing the factor of 2π , or using the wrong variable (x vs y) for the axis of rotation, in surface area problems.
- Using the Cartesian arc-length/surface-area formula on a curve given parametrically (or vice versa) instead of switching to the matching dx/dt , dy/dt version.
- Sign or algebra errors when simplifying $\sqrt{1 + (dy/dx)^2}$ — always simplify carefully under a common denominator before integrating.

You should now be able to

- Recognise when an integral is improper (infinite limit or discontinuity) and evaluate it using a limiting process, stating whether it converges or diverges.
- Find the mean value of a function over a given interval.
- Find the arc length of a curve given in Cartesian or parametric form.
- Find the surface area generated when a curve is rotated about the x -axis or y -axis, in Cartesian or parametric form.